

# The crew plugin: technical reference

The low-level guide, for the people who install, configure and extend crew. A companion overview covers the same ground in plain language. Written 20 September 2026 against crew 0.19.92 on main (commit 621d50dc); the Stop-gate fix (0.19.93) is described as it stands on branch crew-stop-gate-record at b6e8c4bb, in review.

**Design constraint.** A role earns its place only if it buys one of three things: an isolated context window, a restricted tool set, or genuinely independent eyes. Everything else is a file, a command, or the user. Every custom subagent loads the entire `CLAUDE.md` hierarchy at startup, so the roster is kept small and the knowledge is kept in files.

## 1. Layout

| Path                                            | Contents                                                                                                              | Count at 0.19.92 |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------|
| plugin/crew/commands/                           | Slash commands, one Markdown file each with YAML frontmatter.                                                         | 28               |
| plugin/crew/agents/                             | Role definitions: <code>tools:</code> , <code>model:</code> , description, procedure.                                 | 54               |
| plugin/crew/skills/                             | Procedures loaded on match; each a directory with <code>SKILL.md</code> and references.                               | 20               |
| plugin/crew/hooks/hooks.json and hooks/scripts/ | Hook registrations, each event once per shell flavour, and the scripts they run.                                      | 20 registrations |
| plugin/crew/tests/                              | pytest suites, fixtures, and <code>sabotage.py</code> , which reintroduces known bugs and asserts the suite goes red. | 51 modules       |
| plugin/crew/evals/                              | <code>claude plugin eval</code> cases with free graders; runners in <code>scripts/run-plugin-evals.{sh,ps1}</code> .  | 5 cases          |
| plugin/crew/CONFIG.md                           | Every configuration key, its layers and its ratchet rule.                                                             | 18 sections      |

## 2. Roles

### The ladder

| Tier | Roles                                                                                            | Tools of note                                                                                         | Add when                                                                                          |
|------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 0    | explorer, qa-reviewer                                                                            | explorer: Read, Grep, Glob only.<br>qa-reviewer: read-only plus Bash.                                 | Always.                                                                                           |
| 1    | security, smoke-author, developer                                                                | security: no Write or Edit, has Bash. developer: full write, own context, never reviews its own diff. | Security findings reach review; coverage gaps regress; the PM runs work end to end.               |
| 2    | dba, docs-writer, browser-tester, analyst, planner, infrastructure-architect, scribe, researcher | planner works from an abstracted brief; secondOpinion.sendsCode is false by design.                   | Per the table in skills/crew-scaling/SKILL.md; /crew:scale reads .crew/metrics.md and recommends. |
| 3    | parallel sessions, worktrees, agent teams                                                        |                                                                                                       | Every involved repository has green smoke.                                                        |

### The PM

Standing teammate named `crew-pm`, reached by `SendMessage` so it keeps its transcript. Tools: Read, Write, Edit, Bash, Grep, Glob, Agent, Skill. No `SendMessage`, no `ListAgents`, so `resume-by-id` belongs to the caller (`commands/pm.md`). Authority is `report-only | act | autonomous`; the stops in `crew_state.AUTONOMOUS_STOPS` hold at every tier: offboarding a role, deleting a codemap or diagram, rewriting `.crew/metrics.md`, destroying git history. Dispatch cap 3 per pass. Write scope: `.crew/**`, `TODO.md`, `.work/**`, `docs/diagrams/**`, enforced by the write guard from 0.19.92.

### Specialists

`crew_state.SPECIALIST_ROLES` lists 40 roles with no tier, onboarded per repository with `/crew:pm onboard <role>` and justified by the stack (the agent file names the file that proves it). Onboarding a specialist never moves the tier. Groups: languages and frameworks (`python-pro`, `node-developer`, `php-pro`, `rust-engineer`, `react-specialist`, `angular-architect`, `dotnet-core-expert`, `dotnet-framework-4.8-expert`, `sql-pro`, `graphql-architect`, `backend-developer`, `api-designer`); cloud and infrastructure (`terraform-engineer`, `network-engineer`, `platform-engineer`, `microservices-architect`, `legacy-modernizer`, `database-administrator`, `git-workflow-manager`, `workflow-orchestrator`, `multi-agent-coordinator`); Microsoft platform (`windows-infra-admin`, `exchange-online-specialist`, `sharepoint-developer`, `power-automate-specialist`, `powershell-5.1-expert`, `powershell-7-expert`, `powershell-security-hardening`, `ad-security-reviewer`); security and compliance (`penetration-tester`,

compliance-auditor, code-reviewer, architect-reviewer, fintech-engineer, payment-integration); review and writing (qa-researcher, kimi-consult, ai-writing-auditor, skill-author, design-bridge).

## Model routing

`crew_config.py --models` prints the effective provider, model and family per role, which fallbacks are armed, and whether the self-review guard bars anything. The QA order tries Codex (gpt-6-astra) first, then Copilot (Kimi), then Claude; a reviewer of the same family as the diff's author is struck. Config keys `dev.roles.*`, `qa.roles.*`, `dev.fallback`, `qa.fallback`. The Agent tool's model override applies per dispatch.

## 3. Commands

| Command                | Function                                                                                                                                                              | Reads / writes                                                                                                |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| init                   | Phased, resumable setup; each phase asks before writing.                                                                                                              | <code>.crew/config.json</code> , <code>.crew/verify.json</code> , <code>_verify/</code> , CLAUDE.md template. |
| onboard                | Writes the code map; <code>--refresh &lt;subsystem&gt;</code> re-derives one file against the graph.                                                                  | <code>.crew/codemap/*.md</code> with DERIVED/JUDGEMENT markers and <code>anchor: commits</code> .             |
| ticket,<br>work, split | Scope; implement, review, gate end to end; split an oversized Jira ticket.                                                                                            | <code>.work/tickets/</code> , tracker via MCP.                                                                |
| review                 | Hostile QA on the diff: Codex, Copilot or a fresh Claude context, first that probes clean; specialist reviewers by path rule.                                         | <code>.work/review/&lt;name&gt;/{diff,prompt,out}.txt</code> ; a row in <code>.crew/metrics.md</code> .       |
| verify                 | Builds the verification map from evidence; <code>--all</code> runs the whole map with no budget; <code>--price (0.19.93)</code> measures and writes seconds.          | <code>.crew/verify.json</code> (tracked).                                                                     |
| gate                   | Exports, removes and restores branch protection.                                                                                                                      | Bitbucket or GitHub API; export file.                                                                         |
| promote                | Promotes to the next environment; refuses without a <code>.work/PROMOTIONS.md</code> row and, if <code>change.requireForProduction</code> , without a change request. | <code>.work/PROMOTIONS.md</code> .                                                                            |

| Command                                                  | Function                                                                                                                                                                      | Reads / writes                                                                     |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| change                                                   | Files a change request in SDP, Jira or <code>.work/changes/</code> ; refuses while any of the ten template questions is unanswered.                                           | Tracker via MCP.                                                                   |
| emergency                                                | Stands gates down, opens parallel lanes, records what was skipped.                                                                                                            | <code>.crew/incidents/</code> .                                                    |
| debug                                                    | Systematic debugging: read, reproduce, isolate, prove, then fix. Adapted from superpowers (MIT).                                                                              |                                                                                    |
| pm, roster, model, scale, config, upgrade                | Manager conversation; role listing; per-role model set and probe; grow or shrink recommendation; config provenance; schema migration (currently to schema 7).                 | <code>.crew/config.json</code> ,<br><code>~/.claude/crew/config.json</code> .      |
| plan, survey, docs, reference, diagram, runbook, handoff | Second-opinion design; gap survey; docs touched by a change; API/feature reference with anchors; Mermaid diagrams with <code>%% Anchors:</code> ; runbooks; the handoff note. | <code>docs/</code> , <code>docs/diagrams/</code> , <code>.work/HANDOFF.md</code> . |
| jira-sync, sdp-sync, obsidian-sync                       | Two-way ticket sync with the local cache.                                                                                                                                     | MCP servers; Obsidian Kanban file.                                                 |

## 4. Hooks and guards

| Event        | Matcher          | Script                                              | Behaviour                                                                                                                                                                                                                                                           |
|--------------|------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SessionStart | *                | pm-brief, platform sync, context resume             | Injects crew state and any handoff note; corrects <code>platform.os/shell</code> .                                                                                                                                                                                  |
| PreToolUse   | Bash, PowerShell | crew_guards.py via shell shims                      | <code>guards.terraformApply</code> , <code>guards.forcePush</code> , <code>guards.adminMerge</code> (block   ask   allow); <code>guards.prodDatabase</code> , <code>guards.prodServer</code> (none   read   full) against <code>production.databases/hosts</code> . |
| PreToolUse   | Write Edit       | role_write_guard.py via role-write-guard. {sh, ps1} | <code>guards.roleWrites</code> (off by default; block is the floor). See section 6.                                                                                                                                                                                 |

| Event        | Matcher | Script                                      | Behaviour                                                                                              |
|--------------|---------|---------------------------------------------|--------------------------------------------------------------------------------------------------------|
| PreCompact   | *       | handoff writer                              | Writes .work/HANDOFF.md before summarisation.                                                          |
| Notification | *       | pm_pulse.py                                 | Re-briefs the PM with triggers.                                                                        |
| Stop         | *       | verify-gate.<br>{sh,ps1}, context-<br>watch | Runs the matched rules within verify.stopBudgetSeconds (default 60); blocks on failure. See section 7. |

Registration rules: each event registered once per flavour with shell: "powershell" on the Windows entry, because a bare hook command goes to Git Bash on Windows. Every ratcheted key resolves to the narrower of the repository and machine-global layers; a corrupt layer resolves to the strict value; change.requireForProduction ratchets the other way (on wins). A guard that can block ships off and needs a committed must-block/must-allow suite, sabotage-tested.

## 5. Knowledge files

| Artefact   | Location                                             | Freshness rule                                                                                                                                                                                   |
|------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Code map   | .crew/codemap/, un-ignored by name                   | An anchor: behind HEAD means re-check with <code>git diff --name-only &lt;anchor&gt;..HEAD --&lt;paths&gt;</code> ; empty output means current. Comparison is <code>sha[:7] == head[:7]</code> . |
| Code graph | graphify-out/graph.json and GRAPH_REPORT.md, tracked | Rebuilt by post-commit and post-checkout hooks via <code>graphify update .</code> ; a hand build with <code>--no-viz --code-only</code> writes only one of the pair.                             |
| Diagrams   | docs/diagrams/*.mmd with %% Anchors:                 | <code>/crew:diagram refresh</code> rewrites only diagrams whose cited files moved.                                                                                                               |
| Metrics    | .crew/metrics.md                                     | One row per review; grouped by ticket column for the BLOCK+FIX rate.                                                                                                                             |
| Handoff    | .work/HANDOFF.md                                     | Injected at session start until finished or deleted.                                                                                                                                             |

## 6. Changes 0.19.90 to 0.19.92

| Version | Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Files and tests                                                                                                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.19.90 | Foreground execution rule in six role files; PM returns the real dispatch id or "not resumable"; idle signal is recency not liveness, two-cycle escalate rule; write-scope sentences aligned with the guard; qa-reviewer's return stays defect-lines-or-CLEAN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | agents/{developer, pm, python-pro, browser-tester, smoke-author, qa-reviewer}.md, commands/pm.md; tests/test_foreground_execution.py (15 tests). Codex: 5, 4+1, 6, CLEAN.                                               |
| 0.19.91 | Eval suite: pm-does-not-write-code, qa-reviewer-stays-read-only, developer-defers-unrelated-bug, developer-runs-command-in-foreground, pm-answers-status-mid-pass. Structural graders (ticket as clause subject/object with a whole-clause hedge guard; four-field verdict with a numeric reproduction). Runners: comma xfail list, case-sensitive, exit 127 without CLI, zero cases fails, xfail only for a scored failure from this run with no per-run error, passing xfail fails the gate. Workflow tests the secret via a job-level env flag.                                                                                                                                                                                                    | evals/, scripts/run-plugin-evals.{sh, ps1}, .github/workflows/plugin-evals.yml. Live finding: PM case scored 0. Codex: 10; 1+11; 4; 2; 3; 1+1; 1+1.                                                                     |
| 0.19.92 | Role write guard. Policy partition derived from each agent's tools;; test_policy_table_matches_the_agent_files re-derives it. crew_config.layer_state(path) classifies each config layer as absent, ok or corrupt (directory, empty, non-object, guards null/list, dangling symlink); any corrupt layer forces block. Targets resolved component-wise: Windows collapses . . lexically before junctions, POSIX resolves symlinks first; outside-repo and other-drive targets are out of scope and allowed with reason; \\?\\ paths refused for scoped roles; stdin read as bytes with BOM stripped; both shims set PYTHONUTF8 and fail closed on launch failure; outer handler cannot raise; foreign namespaces and unknown roles allowed and logged. | hooks/scripts/role_write_guard.py, role-write-guard.{sh, ps1}, crew_config.py, crew_guards.py, CONFIG.md §18; tests/test_role_write_guard.py (114 cases). Codex: 2+4, 7+2, 3+1, 2, 1, CLEAN; PowerShell specialist 2+2. |

## 7. The Stop gate: 0.19.93 in review

### Root causes, measured 19 September

- verify-gate.sh diffs against .crew/.verify-verified-at, written by record\_verified only when DEFERRED\_COUNT == 0. rules[8] (185 s) against verify.stopBudgetSeconds 60 deferred every turn; the marker sat at dedd1150, 41 commits behind, so 59 files matched 13 rules per Stop. Hot fix applied locally: marker := gated HEAD.

- A rule without seconds is never deferrable. Ten of 21 here (PR #197 priced them); 28 of 28 in TheSelectSource; 43 of 43 in SRL, which also had no marker file and a 499 s Stop.
- In TheSelectSource, rules 0–9 run `_verify/smoke.sh` without `--ci`, whose else-branch runs about 18 checks over SSM against the dev web host on every Stop, with `ENV="${ENV:-dev}"` selecting the target from the ambient environment.
- Exit 77 (SKIP under `smoke.sh`) was a hard FAIL under the gate runner.

## Components on the branch

| Component            | Implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Per-rule record      | <code>verify_record.py</code> : <code>.crew/.verify-gate.record.json</code> , keyed by <code>rule_key()</code> (canonical JSON of paths, run, env, reach, requiresCleanTree; one function for both flavours via the <code>rule-key</code> subcommand). Written atomically; a corrupt or non-dict file poisons the fingerprint so nothing skips; a failed persist withholds both markers; entries whose rule no longer exists are pruned on sync. Sync status starts at 99 and reaches 0 only on a confirmed sync. DEFERRED_COUNT counts acute deferrals only; chronic, reach-excluded, clean-tree and skipped rules stay unverified in the record and are reported every run.                                                                                                                                                                                                                                                                                           |
| Timings              | <code>.crew/.verify-gate.timings.json</code> : <code>max(1, ceil(elapsed))</code> per unpriced command, labelled measured-not-declared; never written to the tracked map by the hook.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>--price</code> | <code>verify_price.py</code> , dispatched only on the literal argument; <code>hooks.json</code> passes none or <code>--all</code> . Applies the gate's env pinning and declared env; refuses any rule the Stop path would defer; records 77 as SKIP with no timing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Reach                | <code>verify_record.scan_reach</code> , reject-only: a verb ( <code>\bssm\b</code> , <code>\bssh\b</code> , <code>\baws\b</code> , <code>\baz\b</code> , <code>\bgh\b</code> , <code>\bpsql\b</code> , <code>\bmysql\b</code> , <code>\bcurl\b</code> and others, word-bounded) anywhere in the command defers; any script, interpreter with a file argument, inline shell ( <code>-c</code> , <code>-Command</code> ) or <code>cd</code> defers with "declare reach: local on rules[N]"; <code>bash -n</code> is parse-only and local; comment-only lines and <code>&lt;# #&gt;</code> blocks are ignored; <code>python -m X</code> names no file. <code>default</code> and <code>always</code> go through the same classification. In progress: split the command on <code>;</code> <code>&amp;&amp;</code> <code>  </code> <code> </code> <code>&amp;</code> and newlines and classify each segment. The wrapper-following recursion from earlier rounds is deleted. |
| Env pinning          | <code>ENV</code> , <code>AWS_PROFILE</code> , <code>AWS_DEFAULT_REGION</code> , <code>KUBECONFIG</code> , <code>TF_WORKSPACE</code> unset per rule unless env is declared; identity includes the declared env so two rules with the same text and different env run separately; every Python-to-shell channel writes LF and every bash read site strips CR.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Exit 77, clean tree  | SKIP is neither pass nor fail, withholds both markers, and deletes the previous fingerprint before any early exit. <code>requiresCleanTree: true</code> defers to <code>--all</code> with a named reason.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>--all</code>   | Whole working tree: <code>git ls-files</code> plus staged deletions, rename sources, HEAD diff and untracked.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Component    | Implementation                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Triggers     | <code>crew_state.read_verify_health</code> : <code>verifyMarkerStale</code> , <code>verifyRulesUnpriced</code> , <code>verifyReachUndeclared</code> ; None on a read failure renders as "unknown" and fires. <code>pm_brief.py</code> gains the three findings.                                                                                                                    |
| Map and docs | This repository's <code>.crew/verify.json</code> declares <code>reach: local</code> on 18 wrapper rules; <code>test_28</code> loads the real map and asserts every rule is declared or plainly local. <code>skills/crew-setup/examples/verify-terraform.json</code> carries seconds and reach. <code>CONFIG.md §18</code> and <code>commands/verify.md</code> describe the record. |

## Review record

| Round      | Findings              | Notable                                                                                                                                                               |
|------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | 7 BLOCK, 5 FIX        | CRLF from native Python broke env unset in bash; exit 77 advanced markers; <code>--all</code> missed staged deletions; corrupt caches did not poison the fingerprint. |
| Specialist | 1 BLOCK, 1 FIX, 1 NIT | Sync status defaulting to 0 advanced markers with no Python present.                                                                                                  |
| 2          | 2 BLOCK, 6 FIX        | Nested wrappers; <code>--price</code> ran wrappers Stop deferred; rule hash omitted env.                                                                              |
| 3          | 2 BLOCK, 3 FIX        | <code>python -m pytest</code> parsed as a wrapper named <code>pytest</code> , which would have deferred eight rules here every Stop; caught by the new self-check.    |
| 4          | 4 BLOCK, 1 FIX        | Three new ways to hide a wrapper from inspection; design inverted to reject-only.                                                                                     |
| 5          | 2 BLOCK               | Compound commands classified whole; fix in progress.                                                                                                                  |

## Operational notes

- Hot fix for an existing repository: `git rev-parse HEAD > .crew/.verify-verified-at`, only if HEAD is a verified state. If it is not, this marks unverified commits as verified and the gate will not look at them again.
- After upgrading, declare `reach` on wrapper rules, run `/crew:verify --price` once, and run `/crew:verify --all` before a PR because a deferred rule is unverified until it runs.
- Open sessions load hooks at start; run `/reload-plugins` to pick up a new version.

## 8. Known items on the fix list

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- `agents/developer.md:178` forbids the commit that every brief asks for; decide once in the role file.
- Developer roles resolve to Claude here although the machine-global config pins Codex; the routing failed silently.
- Every hook's Python writes stderr with the OEM code page when piped on Windows; reconfigure to UTF-8 at each entry point.
- `scripts/check-marketplace.py` crashes on a tracked file deleted but not staged.
- `test_verify_gate_lock_window.py` has a load-sensitive timing assertion that fails under concurrent suites.
- The PM cannot name a subagent, so a PM-dispatched developer is not resumable by id.
- Install `scripts/skip_plugin/*` on reinstall; CLAUDE.md's `git show/autocrlf` sentence is backwards.

Provenance: counts read from the repository at commit 621d50dc; review figures from `.crew/metrics.md`; Stop-gate details from branch `crew-stop-gate-record` at b6e8c4bb plus the in-progress round-5 fix. Not verified: the Stop-gate fix's effect on live sessions; `drift-detection.sh` was not run for any of this week's merges.